Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/02/2004
PI Name:	Becker, Jeanne L. Ph.D.		
Project Title:	3D Growth Effects on Drug Resistance in Human Ovarian Tumor Cells		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	18
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ04HE30G		
Performance Goal No.:	n/a		
Performance Goal Text:	n/a		
Task Description:	One of the problems in studying drug resistance in human cancer is the lack of suitable systems which reproducibly model conditions occurring in vivo. Clinical drug resistance in vivo generally occurs concomitantly with, or shortly thereafter, the administration of chemotherapy and very often manifests as resistance to multiple agents even in only single agent therapy is given. The nature of this rapid onset of drug resistance is poorly understood. The purpose of this research is to culture human ovarian tumor cells three- dimensionally in the NASA-developed High Aspect Ratio Rotating-Wall Vessel (HARV), which provides a model that is more biologically representative of conditions occurring in vivo. This model will be utilized study chemotherapeutic drug resistance developing as a function of three-dimensional growth The task objective of this research is to examine specific cell cycle changes and apoptosis associated with the modulation of human ovarian tumor cell drug resistance during three dimensional cell culture in the		

	HARV. In contrast to acquired resistance, mechanisms underlying the rapid onset of drug resistance in human cancer are not well elucidated. It has been suggested that this type of resistance is not unicellular in nature but is, in fact, multicellular and a function of cell-cell and cell-matrix interactions and three-dimensional architecture. Thus, the development of tumor models using three-dimensional cell culture technology affords great potential not only for improving cancer therapy, but also for providing a superior model for gaining a better understanding of methods to overcome the development of rapid resistance
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The earth based benefits from this research derived from the ability to provide a model more human ovarian tumor growth that is more biologically representative than other currently available models. In vivo, ovarian tumor cells spread from the primary lesion via exfoliation. These peeled off masses of cells float as non-vascularized small aggregates in ascitic fluid, where they subsequently attach to peritoneal surfaces and in doing so disseminate throughout the peritoneal cavity. Growth as three dimensional constructs in the HARV is a particularly relevant model for the study of metastatic human ovarian cancer, in that maintenance of multicellular aggregate structures in a suspended culture environment simulates closely the aspects of both solid and suspension tumor growth that occurs in this disease in vivo.The cells grown in three dimensional constructs exhibit significantly elevated drug resistance, as a function of their three dimensional multicellular architecture, and also exhibit significantly reduced proliferative phase fractions, again mirroring what occurs in large, bulky ovarian tumors in vivo.
Task Progress:	Studies have continued with the OVCAR-3 human ovarian tumor cell line. The cells are resistant to taxol in monolayer, with an ED50 of approximately 200 nM, whereas in 3D HARV culture the cells are resistant to >1 uM of drug. Work this year evaluated the effects of two additional chemotherapeutic agents, cisplatin (which was relatively ineffective up to concentrations exceeding 200 uM) and carboplatin which was somewhat more effective. A large study was conducted which evaluated global gene expression patterns in 3D constructs exposed to carboplatin and taxol, two drugs of choice for treatment of human disease. The constructs were grown for four weeks prior to drug exposure, at which times the cells present in the constructs exhibited high resting phase fractions, and correspondingly low proliferative phase fractions, reflective of the in vivo biologic behavior of bulky, advanced stage human ovarian cancer. The constructs were treated for four days with 200 uM carboplatin, 1 uM taxol, or a combination of both agents; controls received vehicle. For each group N=10, thus affording a strong base for evaluation of statistically significant differences in gene expression patterns between treatments. Following drug exposure, assays of cell cycle analysis, cell viability and apoptosis were performed, and RNA was harvested and microarray analysis was carried out for 7,820 genes. Data are currently under evaluation, but early analysis reveals significant differences between treatments for a variety of gene groups, notably alpha and beta tubulins. Utilization of this three-dimensional model to evaluate global gene expression in response to chemotherapeutics agents provides much insight into targeted molecular approaches to overcoming drug resistance in this disease. Work on LN-1 human ovarian tumor cells cultured aboard ISS during expedition 3 has also continued. Cytokine production by LN-1 aggregates grown during the 14 day study has been evaluated as has antigenic proteins preserved in an RNA stabilizing agent, demonstrating a new methodology for obtaining protein data from these archived specimens. These studies have been conducted in conjunction with scientists at Johnson Space Center, and have resulted in the presentation of two abstracts at national meetings. Of note this year is the relocation of the laboratory. The first relocation occurred at the University of South Florida (Tampa, FL) in September 2003. A larger scale move occurred in March of 2004 when I relocated to Houston, TX to assume the position of Associate Director of the National Space Biomedical Research Institute, and I also assumed a faculty position in the Department of Obstetrics & Gynecology at Baylor College of Medicine, where my research now continues. The moves have, unfortunately, resulted in significant “down time” for the laboratory, but productivity has continued in spite of this.
Bibliography Type:	Description: (Last Updated: 08/09/2012)
Abstracts for Journals and Proceedings	Becker JL, Widen RH. "Inhibition of OVCAR-3 growth by aspirin and paclitaxel." Society of Gynecologic Investigation 51st Annual Meeting, March 2004. Society of Gynecologic Investigation 51st Annual Meeting, March 2004. Abstract D133. , Mar-2004
Abstracts for Journals and Proceedings	Hammond DK, Becker JL, Elliott TF, Holubec K, Baker TL, Love JE. "Cell growth and cytokine production of human mixed mullerian tumor (LN-1) cells grown on the International Space Station (ISS)." Society for In Vitro Biology World Congress, San Francisco, CA, May 22-26, 2004. World Congress on In Vitro Biology. Abstract Book. May 2004. Abstract VT-2019, p. 80-A. , May-2004